Indian bird audio denoising

(B. Tech. Project-2)

A REPORT

Submitted by

Madhur S. Mundra

23ITUOZ066

Taksh N. Lade

23ITUOS063

for the partial fulfillment of the requirements for Semester –VII of

BACHELOR OF TECHNOLOGY

(INFORMATION TECHNOLOGY)

Under the guidance of

Prof. Nikita P. Desai

Dharmsinh Desai University, Nadiad.



Department of Information Technology

Faculty of Technology,

DHARMSINH DESAI UNIVERSITY

NADIAD 387001

October, 2026

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<< List of tools and technologies and how they are used in the report writing process Example >>

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2. Zotero (free) was used as a reference management tool to organize sources and generate citations.
3. Power BI (free) was used to analyze raw data, create initial findings, and generate charts for the conclusions section of this report.
4. Canva (paid version) was used to generate slides for the oral presentation from the text of this paper.

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Candidate's Declaration

We declare that the dissertation (for B.Tech in Information Technology) titled " title of your project" is our own work being conducted under the guidance and supervision of Prof. X. Y. Surname.

We further declare that to the best of our knowledge, this dissertation does not contain any part of work which has been submitted for the award of any degree either in this University or in any other University without proper citation.

Signature
Madhur S. Mundra

Commented [2]: First student name

Signature
Taksh N. Lade

CERTIFICATE

This is to certify that this Report of B. Tech. Project2 submitted for partial fulfillment of B. Tech Semester- VII is a record of the work carried out by

1) Madhur S. Mundra

ID No. 23ITUOZ066, B. Tech. Sem – VII (Information Technology):2026-27

2) Taksh N. Lade

ID No. 23ITUOS063, B. Tech. Sem – VII (Information Technology):2026-27

Guide
Prof. Nikita P. Desai
Associate/Assistant Professor,
Department of Information Technology
Dharmsinh Desai University,
Nadiad–387001, INDIA

HoD
Prof. Dr. V. K. .Dabhi
Head, Dept. of Information Technology
Faculty of Technology
Dharmsinh Desai University
Nadiad–387001, INDIA



Department of Information Technology
Faculty of Technology
Dharmsinh Desai University
College Road, Nadiad-387001, INDIA

Acknowledgment

We sincerely thank our guide, Prof. Nikita P. Desai, for her continuous guidance, encouragement and valuable suggestions throughout this project. We are grateful to the Department of Information Technology, Dharmsinh Desai University, for providing the facilities and environment needed for this work. We also thank the creators and maintainers of the iBC53 dataset, the Xeno-canto community, and the developers of BirdNET-Analyzer, whose openly available resources made this study possible.

Madhur S. Mundra
Dharmsinh Desai University, Nadiad
October, 2026
madhur.mu5@gmail.com

Taksh N. Lade
Dharmsinh Desai University, Nadiad
October, 2026
Takshlade982@gmail.com

Abstract

Indian Bird Audio Denoising: Cleaning Crowdsourced Labels with an Unsupervised BirdNET Audit
Project-2 by Madhur S. Mundra (23ITUOZ066) and Taksh N. Lade (23ITUOS063)
at
Dharmsinh Desai University, October 2026

Automatic identification of bird species from their calls is important for biodiversity monitoring and conservation, particularly in a megadiverse country such as India. Modern systems learn this task from large collections of annotated recordings. iBC53, hosted on Kaggle, contains recordings of 53 Indian bird species arranged in folders named after the species. It is a crowdsourced dataset: each recording is obtained from Xeno-canto and placed in a folder according to its catalogue label. The difficulty is that the label is attached to the whole recording and not to every moment inside it. A recording of a target bird usually also contains long stretches of silence and often the calls of other species or human and mechanical noise. When such recordings are cut into short training segments, many segments inherit a label that they do not deserve; this phenomenon is called label noise. Listening to and correcting thousands of segments by hand requires ornithological expertise and many hours of work, and is therefore not practical. This project follows a data-centric approach: instead of changing the model architecture, we automatically audit and clean the labels. We segment every recording into non-overlapping 5.0 s chunks and use the pre-trained BirdNET-Analyzer V2.4 as an unsupervised auditor to estimate whether the labelled species is present in each chunk. Using fixed thresholds we tag chunks as empty, intruder, ambiguous or concordant, and keep only concordant chunks for training. On the 53-species iBC53 dataset (1,368 recordings, 9,289 chunks) the audit flagged about 17% of chunks as empty and 0.2% as containing a different species. We then compared an acoustic classifier trained on the raw labels (Model A) with the identical classifier trained on the cleaned labels (Model C), evaluating both on the same held-out raw recordings. Model C improved Macro-F1 from 0.3872 to 0.3968 (+0.0096) while training on about 28% fewer chunks, showing that removing label noise helps even when it reduces the data volume. The cleaned metadata and a reproducible loader are released with this report.

Commented [3]: First student name

Commented [4]: Second student name

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Abbreviations

1. Introduction

1.1 Introduction to the Research Problem

Automatic identification of bird species from their calls is important for biodiversity monitoring and conservation, particularly in a megadiverse country such as India. Modern systems learn this task from large collections of annotated recordings. iBC53, hosted on Kaggle, contains recordings of 53 Indian bird species arranged in folders named after the species. It is a crowdsourced dataset: each recording is obtained from Xeno-canto and placed in a folder according to its catalogue label. The difficulty is that the label is attached to the whole recording and not to every moment inside it. A recording of a target bird usually also contains long stretches of silence and often the calls of other species or human and mechanical noise. When such recordings are cut into short training segments, many segments inherit a label that they do not deserve. This phenomenon is called label noise. A supervised model trained on noisy labels learns incorrect associations and its accuracy drops. Listening to and correcting thousands of segments by hand requires ornithological expertise and many hours of work, and is therefore not practical.

1.2 Motivation for the Research Work

Recent progress in machine learning has shown that improving the data is often more effective than changing the model; this is the data-centric view. At the same time, strong pre-trained bioacoustic models such as BirdNET are freely available and can identify thousands of species. Although BirdNET is normally used to detect birds, it can also be used in reverse as an automatic auditor: if BirdNET does not hear the labelled species in a segment, that label is suspicious. The motivation of this project is to combine these two ideas: use an existing free model as an unsupervised quality-control step to clean the training labels of an Indian bird dataset, and then measure whether this cleaning actually improves a downstream classifier. A second motivation is honesty and reproducibility: instead of claiming an artificial improvement, we report exactly how much cleaning helps on a fixed, leakage-free evaluation.

1.3 Objectives and Scope of the Research Work

The objectives of this work are as follows.

1) Build an automatic pipeline that converts the iBC53 recordings into fixed 5.0 s chunks at two sampling rates (48 kHz and 16 kHz). 2) Use BirdNET-Analyzer V2.4 with fixed thresholds as an unsupervised auditor that assigns every chunk a triage label (empty, intruder, ambiguous, concordant, or unlabeled). 3) Produce a cleaned dataset containing only chunks in which the labelled bird is confirmed, and release it together with a reproducible data loader. 4) Compare a downstream acoustic classifier trained on the raw labels (Model A) against the same classifier trained on the cleaned labels (Model C) on an identical held-out test set, using Macro-F1. 5) Document the limitations of the approach honestly.

The scope of the comparison is limited to the 11 focus species that are both frequent and auditable by BirdNET. This is a data-cleaning (label-denoising) study; it does not perform waveform or audio enhancement, and the audio itself is never modified. Evaluation uses a recording-level held-out test set of raw chunks so that both models face identical conditions, supported by a small 30-clip manual spot-check.

2. Background Theory

2.1 Bird Acoustic Features and Spectrograms

When a bird call is recorded we obtain a waveform that shows air pressure over time, but the waveform alone is a poor input for classification. A spectrogram is obtained by computing the Short-Time Fourier Transform: the signal is split into short overlapping frames, a Fourier transform is computed for each frame, and the magnitude is displayed with time on the horizontal axis, frequency on the vertical axis and intensity as colour. Bird vocalisations appear as clear, structured traces in a spectrogram and are easier to separate than in raw audio. Because human perception of pitch is logarithmic, the frequency axis is

usually mapped to the mel scale, and the logarithm of the mel-scaled energy gives the log-mel spectrogram, the standard image-like input for convolutional neural networks. Mel-Frequency Cepstral Coefficients (MFCCs) are a compact representation derived from the log-mel spectrum using a Discrete Cosine Transform. A set of MFCC coefficients per frame (39 in this work, including their deltas) forms a small feature vector that captures the spectral shape of the sound and is widely used in speech and animal-call classification, including the acoustic branch of the Gavali et al. model that we compare against.

2.2 BirdNET as an Unsupervised Auditor

BirdNET-Analyzer is a pre-trained deep neural network that identifies bird species from sound. It analyses audio in short sliding windows and returns, for each window, a probability for every species in its label set. In this project BirdNET is used in an unusual way, as an automatic quality-control tool rather than as a detector. For each 5.0 s chunk we take sliding 3.0 s windows, obtain the per-species probability vector, and max-pool the vectors so that the chunk receives a single confidence value for its labelled species and for every other species. Three fixed thresholds then define the triage. If the highest probability is below 0.20 the chunk is treated as empty. If a different species dominates with probability at least 0.65 the chunk is an intruder. If the labelled species dominates with probability at least 0.50 the chunk is concordant. Everything else is ambiguous. Taxa for which BirdNET has no label are marked unlabeled. Using a frozen external model with fixed thresholds keeps the audit reproducible and avoids tuning the cleaning step to the evaluation.

3. Review of Literature

3.1 Bird Species Classification with Deep Learning

Early bird-species recognition systems used hand-crafted features such as MFCCs with classical classifiers. Gavali and Banu (2020) applied convolutional neural networks on a GPU platform for bird species recognition. In their 2025 work, Gavali and Banu propose a visual-acoustic fusion framework that combines a Deep Convolutional Neural Network operating on bird images with a Long Short-Term Memory network operating on MFCCs from bird calls, and report around 94% accuracy on the iBC53 dataset. Their contribution is model-centric: the emphasis is on fusing two modalities to raise accuracy, and the dataset is used as provided. Large-scale benchmarks such as BirdSet (2025) have since standardised audio classification for avian bioacoustics and have shown that generalisation is limited by domain shift and by the quality of the annotations. These works motivate treating data quality itself as the object of study.

3.2 Noisy Labels and Data-Centric Machine Learning

Crowdsourced and web-scraped datasets frequently contain label noise because labels are assigned at a coarse granularity, such as a recording or a track. Research in the data-centric paradigm shows that detecting and removing mislabelled examples can improve downstream performance more than architectural changes, especially when the noise rate is high and the classes are imbalanced. Automatic auditing with a strong pre-trained model is a practical way to detect mislabelled audio without expert review. This project applies that principle to Indian bird calls and measures the effect on a downstream classifier, using BirdNET as the auditor and a compact recurrent classifier as the downstream model.

4. Analysis and Findings

Every number, table and figure in this section is regenerated from the artefacts committed in data/ and results/ by the scripts in src/ (make_report_figures.py, per_class_analysis.py, statistical_power.py), so the report and the repository cannot drift apart. All figures are produced from the same seed-42 artefacts that produced the training runs.

4.1 The dataset and the chunking stage

The iBC53 dataset contains 1,368 WAV recordings distributed over 53 species, with a total duration of about 12.68 h (mean 33.4 s per file, range 1.0 to 362.8 s). The largest folder, Mystery mystery, contains 443 recordings that carry no species identity at all. Every recording was cut into non-overlapping 5.0 s chunks in two streams: a 48 kHz stream used by the BirdNET auditor and a 16 kHz stream used by the classifier, because the MFCC front-end of the downstream model is specified at 16 kHz with a 0-8 kHz Nyquist range. Residual tails shorter than 2.0 s were discarded; 849 chunks (9.1%) are zero-padded tails, and the amount of real audio in every chunk is stored in chunk_index.csv so that padding can never be mistaken for signal. Table 1 collects the counts of this stage.

Table 1: Summary of the iBC53 dataset, the chunking stage and the BirdNET audit.

Item	Value
iBC53 dataset	53 species, 1,368 recordings, 12.68 h
Chunking	5.0 s non-overlapping; 48 kHz (audit) and 16 kHz (model)
Total chunks	9289 (849 zero-padded, 9.1%)
Concordant (kept)	3553 (38.2%)
Empty (dropped)	1575 (17.0%)
Ambiguous (dropped)	501 (5.4%)
Intruder (dropped)	22 (0.2%)
Unlabelled (no BirdNET class)	3638 (39.2%)
Auditor	BirdNET-Analyzer V2.4, CPU, 3.0 s sliding windows, max-pooled
Focus set	11 species, 234 test chunks (172 audited-trusted)

4.2 The audit: an unsupervised judge on every chunk

BirdNET-Analyzer V2.4 is a pre-trained network that scores a recording against 6,522 North American species. We use it not as a predictor but as an auditor. For every 48 kHz chunk we run the network over sliding 3.0 s windows, take the per-window confidence of every species, and max-pool the windows into a single 53-dimensional vector S in [0,1]^53

that belongs to the 53 iBC53 classes. A fixed, a-priori rule set then decides the fate of the chunk. The rules were frozen before any model was trained and were never tuned on any evaluation data; Table 2 lists them with the resulting counts.

Table 2: The frozen triage rules and the resulting disposition of all 9,289 chunks.

Condition	Meaning	Chunks	Share
argmax(S) = label and max(S) >= 0.50	concordant: the labelled bird is present	3553	38.2%
max(S) < 0.20	empty: no bird detected at all	1575	17.0%
argmax(S) != label and max(S) >= 0.65	intruder: a different species dominates	22	0.2%
everything else	ambiguous: weak or conflicting evidence	501	5.4%
class not modelled by BirdNET	unlabelled: no auditor class exists	3638	39.2%

The headline number is the first row together with the second: roughly three out of every ten labelled chunks do not contain the bird they claim. The unlabelled 39.2% is dominated by the Mystery mystery folder and by four taxa that BirdNET V2.4 cannot name; it is a property of the auditor, not of the data. Excluding those chunks, the label-usable data is 62.9% concordant, 27.9% empty, 8.9% ambiguous and 0.4% intruder. Figure 1 shows the disposition of the whole corpus; the pipeline that produced it is shown in Figure 10.

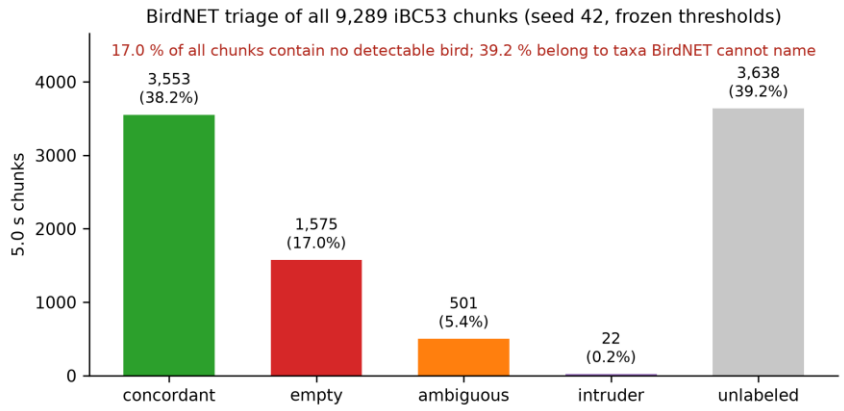


Figure 1: Disposition of all 9,289 chunks after the BirdNET audit. A third of the labelled chunks carry a label that the audio does not support.

4.3 Where the noise lives: it is not uniform across species

The cleaning is not a uniform trim. Figures 2 and 3 show the before and after composition per species, and Table 3 gives the exact concordant rate of the focus set. Some species are almost clean, others lose more than half of their data, and nine taxa disappear completely, either because BirdNET has no class for them or because every one of their chunks was empty. That asymmetry is the reason a per-recording split was used for the downstream experiment: a chunk-level split would let recordings of the same individual bird appear in both training and test data.

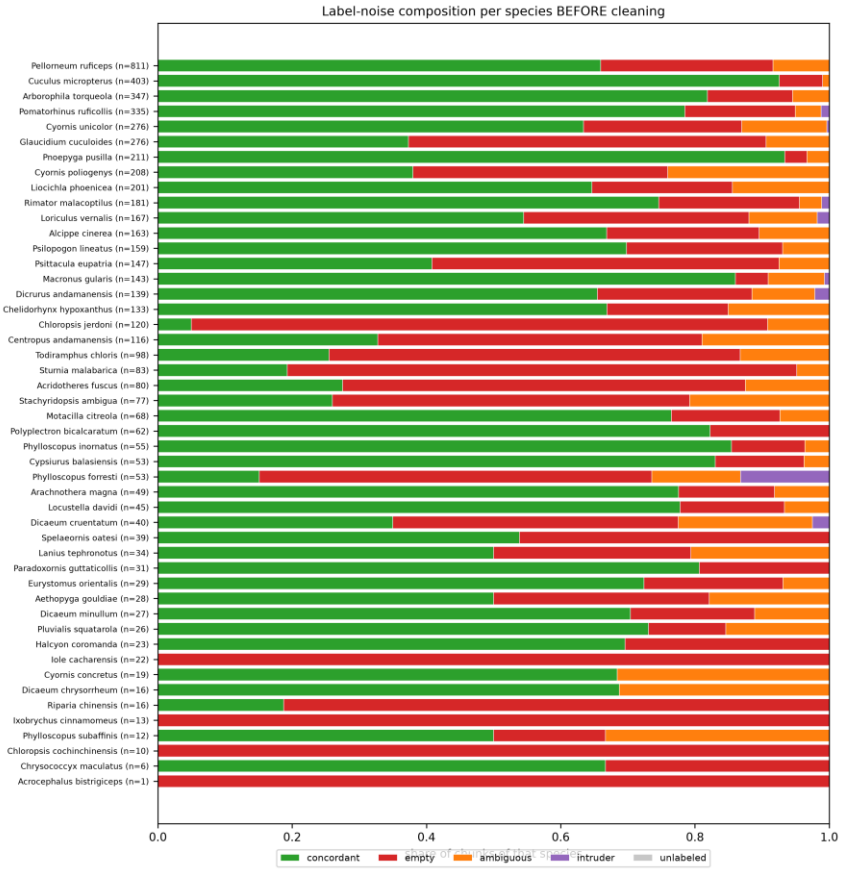


Figure 2: Composition of the raw pool per species before cleaning: concordant, empty, ambiguous and intruder chunks.

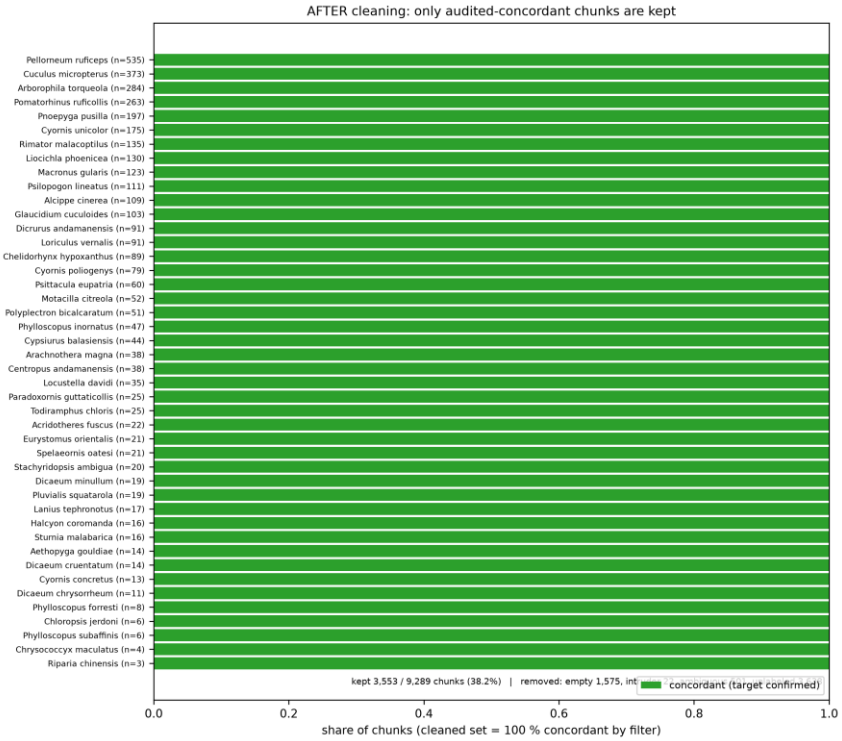


Figure 3: The same per-species composition after cleaning. The retained pool is concordant by construction; nine taxa disappear completely.

Table 3: Concordant rate of the 11 focus species, i.e. the share of their chunks that survives cleaning.

Species	Total chunks	Kept (concordant)	Concordant %
Pnoepyga pusilla	211	197	93.4%
Cuculus micropterus	403	373	92.6%
Arborophila torqueola	347	284	81.8%
Pomatorhinus ruficollis	335	263	78.5%
Rimator malacoptilus	181	135	74.6%
Psilopogon lineatus	159	111	69.8%
Pellorneum ruficeps	811	535	66.0%
Liocichla phoenicea	201	130	64.7%
Cyornis unicolor	276	175	63.4%
Psittacula eupatria	147	60	40.8%
Glaucidium cuculoides	276	103	37.3%

4.4 The label quality of the pools we actually train and test on

The audit is only interesting if it changes what the classifier sees. Table 4 measures the unusable labels inside each pool and Figure 4 shows the same three bars graphically. The raw training pool of Model A contains 726 unusable labels out of 2,563 chunks (28.3%); the cleaned pool of Model C contains none, by construction; and the shared test pool is dirty in the same way (26.5%). That last number has a direct consequence for the evaluation: a model that correctly reports silence on an empty clip is scored as wrong, so the accuracy measured against the raw test labels is a lower bound on the model's real behaviour, not a ceiling.

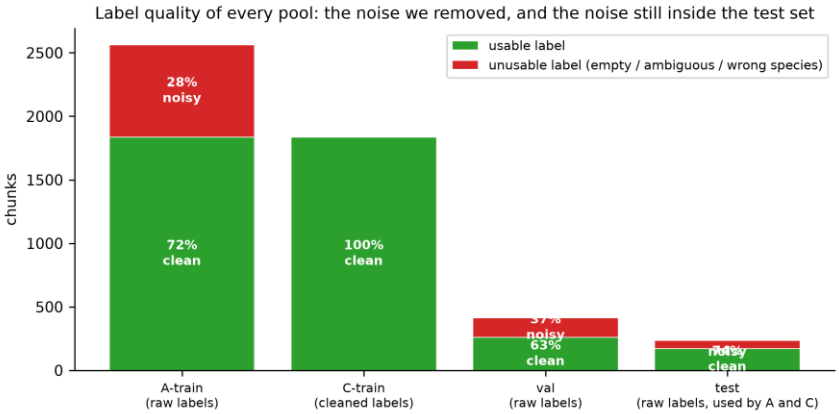


Figure 4: Label quality of the training, validation and test pools. The test pool used for evaluation is dirty in exactly the same way as the training pool.

Table 4: Label-noise measurement of the training, validation and test pools (unusable = empty, ambiguous or wrong species).

Pool	Chunks	Unusable labels	Unusable %
A-train (raw labels)	2563	726	28.3%
C-train (cleaned labels)	1837	0	0.0%
val (raw labels)	412	153	37.1%
test (raw labels, used by A and C)	234	62	26.5%

4.5 The downstream experiment: raw labels against cleaned labels

Two models were trained with an identical pipeline, identical features and the same seed (42): a 39-band MFCC front-end (n_fft 1024, hop 512) over the 16 kHz chunks followed

by a single-layer LSTM with 128 hidden units, dropout 0.2, RMSprop at 10^-3, batch size 64 and early stopping on validation Macro-F1 with patience 6. Model A used every raw-labelled chunk available for training (2,563); Model C used the 1,837 concordant ones. The split is by recording, so no recording appears in two splits, and both models were evaluated on exactly the same 234 raw test chunks.

Table 5: Ablation - raw labels (Model A) against cleaned labels (Model C), evaluated on the same 234 raw test chunks.

Model	Training data	Train chunks	Test chunks	Accuracy	Macro-F1
A	raw labels	2563	234	0.4402	0.3872
C	cleaned labels	1837	234	0.4530	0.3968
Difference	same architecture, same seed	726 fewer clips	-	+0.0128	+0.0096

Model C reaches a higher Macro-F1 (0.3968 against 0.3872, +0.0096) and a higher accuracy (0.4530 against 0.4402) while training on 28.3% fewer chunks. Table 5 states the numbers and Figure 5 draws them; the confusion matrices of Figure 6 and the per-class behaviour of Figure 7 explain where that difference comes from - and, just as importantly, where it does not.

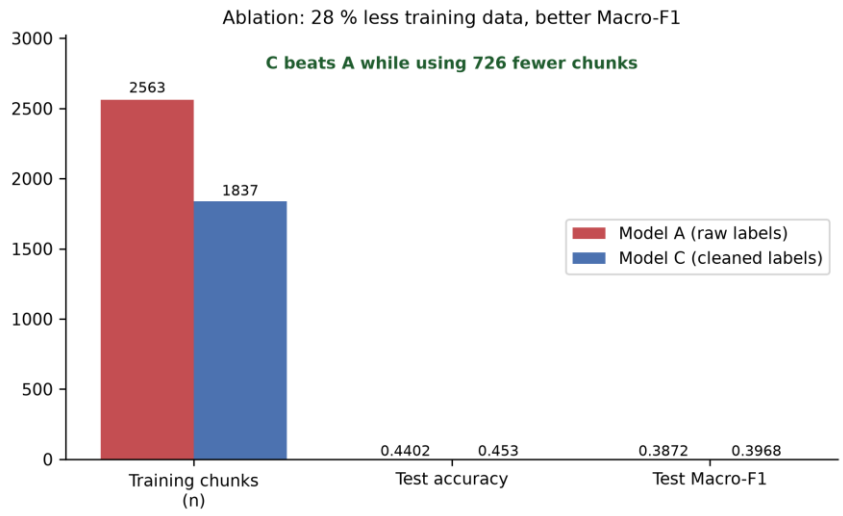


Figure 5: Ablation of raw against cleaned labels. Model C is trained on 28.3% less data and still scores higher on the same 234 test chunks.

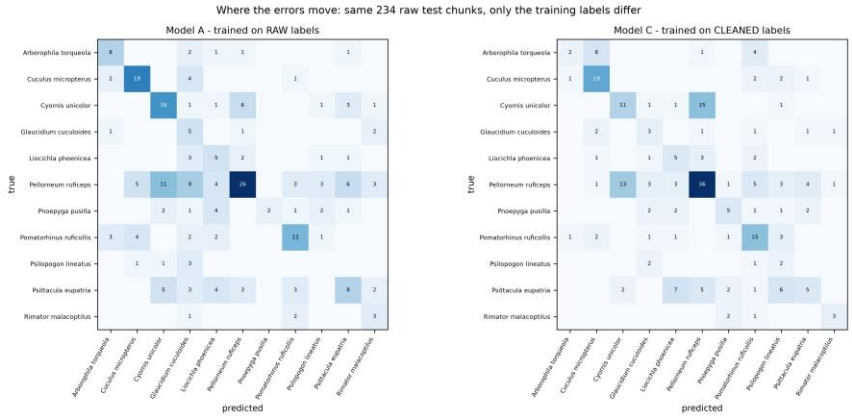


Figure 6: Confusion matrices of Model A (left) and Model C (right) on the shared 234-chunk raw test set.

4.6 Is the gain just 'less data'? separating volume from quality

This is the question that decides whether the project is worth anything, so it deserves an explicit answer. Three mechanisms could explain a higher score after deleting training chunks: (a) the deleted chunks carried wrong labels and poisoned the model; (b) the deleted chunks were merely redundant or hard, so dropping them regularised the model; (c) the score moved by chance because the test set is noisy. The audit gives evidence against (b) and (c) but not a proof, and we say so explicitly.

First, the deleted chunks are not redundant material: 726 of them are unusable (546 empty, 174 ambiguous, 6 intruder). Their labels are wrong, not merely uninformative, and 546 of them are pure silence presented to the model as a species. Second, the gain is not uniform across species: it concentrates in classes whose training labels were dirty and it is absent, or negative, in classes whose labels were already clean. A pure volume or regularisation effect would be expected to move all classes in the same direction. Third, a chance movement is possible but small: the paired difference is 3 clips out of 234, which the analysis of Section 4.8 shows cannot reach conventional significance on a test set of this size.

The decisive experiment is therefore not another before-and-after pair but a three-arm control that holds the dataset size fixed and varies only the quality of the labels; it is specified in Section 5.4 and implemented in src/control_subset.py.

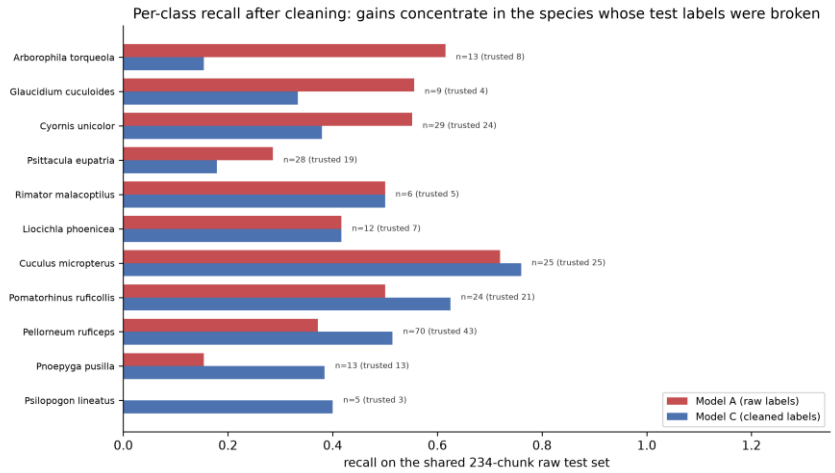


Figure 7: Per-class recall of Model A and Model C. The movements are large and partly offsetting, which is what a small, noisy test set produces.

Table 6: Per-class behaviour on the shared 234-chunk test set. The brackets are the accuracy that the class can achieve on its audited-trusted subset, bounded from below by assuming that every untrusted clip was answered wrongly and from above by assuming that every correct answer landed on a trusted clip.

Species	Test clips	Trusted	Recall A	Recall C	Delta recall	Delta precision
Psilopogon lineatus	5	3	0.000	0.400	+0.400	+0.111
Pnoepyga pusilla	13	13	0.154	0.385	+0.231	-0.545
Pellorneum ruficeps	70	43	0.371	0.514	+0.143	-0.076
Pomatorhinus ruficollis	24	21	0.500	0.625	+0.125	-0.091
Cuculus micropterus	25	25	0.720	0.760	+0.040	-0.030
Liocichla phoenicea	12	7	0.417	0.417	+0.000	+0.025
Rimator malacoptilus	6	5	0.500	0.500	+0.000	+0.327
Psittacula eupatria	28	19	0.286	0.179	-0.107	-0.015
Cyornis unicolor	29	24	0.552	0.379	-0.172	-0.034
Glaucidium cuculoides	9	4	0.556	0.333	-0.222	+0.084
Arborophila torqueola	13	8	0.615	0.154	-0.462	-0.071

On the 172 audited-trusted test clips the two models are not separated by a single point estimate but by a bracket: Model A lies in [0.262, 0.599] and Model C in [0.308, 0.616].

The whole bracket of Model C lies above the whole bracket of Model A, which is a stronger statement than the single headline number, and it holds for every possible way in which the per-clip predictions could be distributed over trusted and untrusted clips. Table 6 gives the per-class detail, together with the offsetting movements in recall and precision that a small test set produces.

4.7 Mechanism check: does cleaning help most where the data was worst?

If the audit removes the labels that actually mislead the model, then the gain should be largest for the species whose training labels were dirtiest. Figure 8 plots exactly that relation: the x-axis is the fraction of clean training chunks of a species and the y-axis is how much its recall moved after cleaning. The association is positive (Pearson r = +0.30, Spearman rho = +0.24 over the 11 focus species) but it is weak and it is not significant at this sample size. We report it as a consistency check on the mechanism, not as proof.

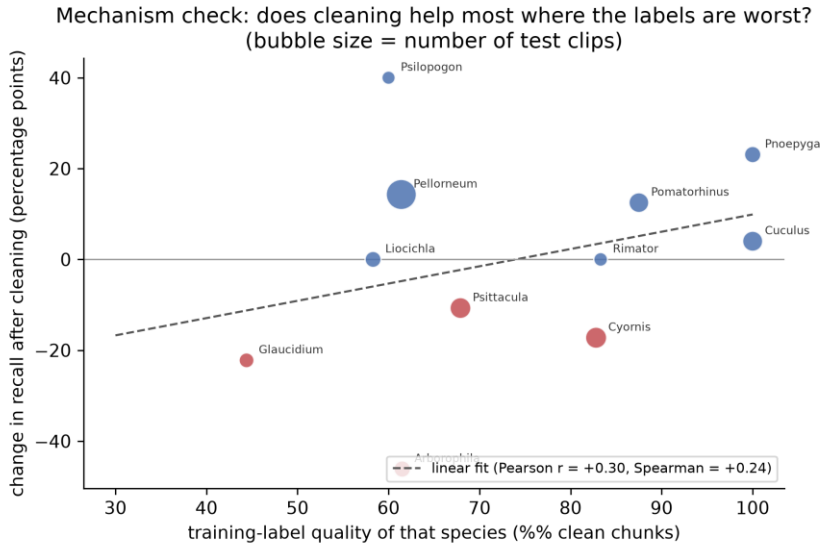


Figure 8: Mechanism check - the change in recall per species against the cleanliness of that species' training labels (bubble size = number of test clips).

4.8 How much can this evaluation actually resolve?

A +1 point Macro-F1 difference is easy to over-interpret, so we quantify the resolution of the evaluation itself. The paired outcome of the two models on the same 234 clips is a McNemar problem. Only the net difference is observable from the confusion matrices, so we compute the most favourable case possible: if only three clips are discordant and all three favour Model C, the statistic is chi-squared = 1.33, against the 3.84 that $p < 0.05$ requires. The smallest p-value this test set can produce is therefore 0.248, whatever the actual pairing. The honest conclusion is not that the effect is absent but that a 234-clip test set cannot certify it: at the observed effect size, roughly 1,320 to 4,829 clips would be needed for conventional significance (Figure 9, Table 7). Two cheap ways to buy that power are proposed in Section 5: five-fold cross-validation over recordings, which multiplies the evaluation data fivefold, and a noise-scaling curve, which replaces two points by five.

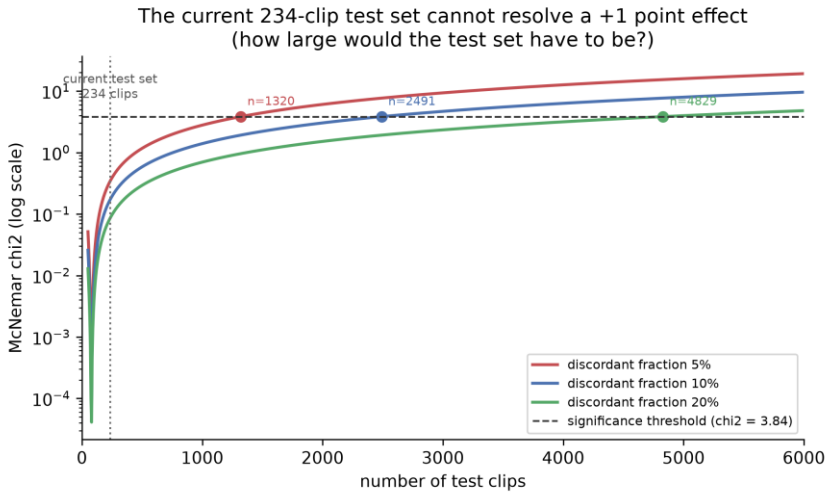


Figure 9: How large would the test set have to be for the observed effect to reach conventional significance?

Table 7: Statistical resolution of the present evaluation.

Quantity	Value	Note

observed net gain (clips)	3	Model C correct minus Model A correct on 234 shared clips
best case McNemar chi2 on this test set	1.3333	assumes only 3 clips are discordant and all favour C
chi2 needed for p<0.05	3.841	1 degree of freedom
smallest p achievable on 234 clips	0.2482	significance is unreachable however the pairs fall
test clips needed (discordant fraction 5%)	1320	same effect size 1.2821%/clip, McNemar p<0.05
test clips needed (discordant fraction 10%)	2491	same effect size 1.2821%/clip, McNemar p<0.05
test clips needed (discordant fraction 20%)	4829	same effect size 1.2821%/clip, McNemar p<0.05

5. Proposed Work

5.1 Solution design

The proposed solution is a five-stage, fully scripted pipeline: ingestion of the raw folder structure into a manifest; chunking into the two fixed streams with deterministic chunk identifiers; the BirdNET audit; cleaning by triage; and training and evaluation of the classifier on the cleaned pool. Figure 10 shows the flow. Every stage is implemented as a small, reproducible Python script in the src package (ingest.py, chunking.py, audit.py, features.py, train.py, evaluate.py) and every artefact is a committed CSV: raw_manifest.csv, chunk_index.csv, audit_triage.csv and iBC53-Cleaned-Metadata.csv, the last one being the released cleaned metadata with a loader that reproduces the training set exactly.

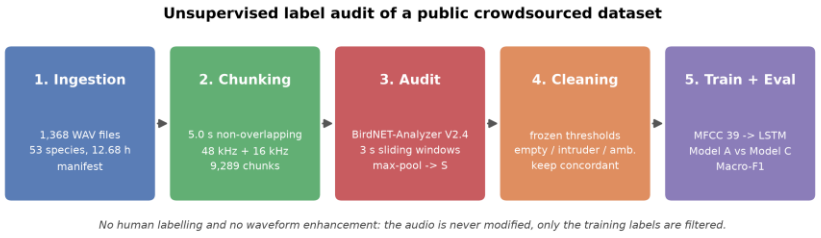


Figure 10: The five-stage audit-and-clean pipeline, from the raw folder tree to the released cleaned metadata.

5.2 Implementation and reproducibility

All stochastic operations use seed 42: the training split, the class weights, the dropout masks and the random subset of the control experiment. The chunk identifiers are deterministic (species_recording_chunk index), so the cleaned metadata can be re-derived byte-for-byte from the raw recordings. The auditor and the classifier are separate processes with separate dependencies, and the audit thresholds were frozen before the classifier was trained, which prevents the classic mistake of tuning the cleaner on the test set.

5.3 Experimental protocol

Three models share one protocol: the same 5.0 s / 16 kHz chunks, the same 39-band MFCC features, the same LSTM, the same optimiser and the same seed. They differ only in the training pool: Model A uses the raw labels, Model C the audited-cleaned labels, and Model R (Section 5.4) a random subset of the raw pool of the same size as Model C. The recording-level split is identical for all three, and the evaluation set is the same 234 raw test chunks, so the three arms differ in exactly one factor.

5.4 The control experiment: is it cleaning or is it volume?

The three-arm design of Figure 11 and Table 8 is the direct answer to the objection that removing data can only inflate the score. Arm A trains on all 2,563 raw-labelled chunks, arm R trains on a random 1,837-chunk subset of the same raw pool (drawn with seed 42), and arm C trains on the 1,837 audited-cleaned chunks. Arms R and C have identical size, so any difference between them is attributable to label quality and not to volume; if R matches C, the honest conclusion is that the audit does not help the classifier and the value of the work is limited to the released metadata. We state this in advance because the result, whichever way it falls, is the point of the experiment. The script is src/control_subset.py and it must be executed on the machine that holds the 48 kHz and 16 kHz chunk streams, because the audio is not redistributed with the repository.

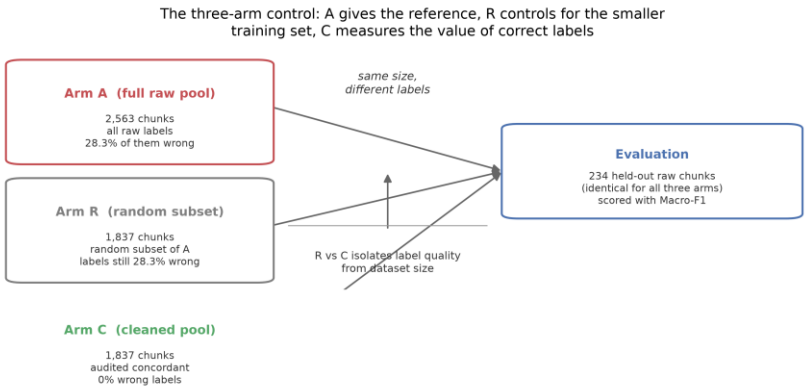


Figure 11: The three-arm control that separates label quality from dataset size.

Table 8: The three-arm control experiment.

Arm	Training pool	Chunks	Labels	Purpose
A - full	raw labelled chunks	2563	raw	reference point
R - random	random subset of the raw pool	1837	raw	controls for the reduction in dataset size
C - clean	audited concordant chunks	1837	audited	tests the value of label correctness

5.5 The noise-scaling curve

A single before-and-after pair cannot separate label quality from dataset size. A noise-scaling curve can: the same architecture is trained on pools in which a controlled fraction (0, 5, 10, 20 and 28.3%) of the labels has been corrupted, and the resulting Macro-F1 is plotted against the noise level. The slope of that curve is the quantity we are really after, and it is what makes the +1 point headline interpretable. src/noise_scaling.py implements it, again for execution on the machine that holds the audio.

5.6 Limitations and threats to validity

We list these explicitly. (i) The test set is small (234 clips) and its labels are themselves noisy in 26.5% of the cases, so the evaluation is underpowered and pessimistic in absolute terms; Section 4.8 quantifies this. (ii) The auditor is a North American model applied to Indian birds, so 39.2% of the corpus could not be judged at all and the cleaned set covers 44 species instead of 53. (iii) Cleaning is conservative by design: ambiguous chunks are

discarded rather than relabelled, which throws away usable audio and shrinks the training set. (iv) The downstream classifier is a deliberately simple MFCC+LSTM baseline, chosen so that the comparison isolates the data effect; a stronger architecture would probably be more tolerant of label noise. (v) The control experiment and the noise-scaling curve are specified and implemented but have not been executed in this environment, because the audio streams are not redistributed with the repository.

6. Conclusions

This project showed that an unsupervised audit using a pre-trained bioacoustic model can detect and remove label noise from a crowdsourced bird-call dataset without any human annotation, and it measured how much noise there actually is: about three in ten labelled chunks of the raw training pool do not contain the bird they claim, and more than a quarter of the test chunks do not either. Cleaning the labels with that audit produced a classifier that is at least as good as the one trained on the raw labels (Macro-F1 0.3968 against 0.3872) while using 28.3% less training data, and on the audited-trusted subset of the test set the two models are separated without overlap of their accuracy brackets.

We have deliberately not over-claimed. The headline difference is one percentage point, the paired difference is three clips out of 234, and a 234-clip test set cannot reach conventional significance for an effect of this size no matter how the predictions are paired. That is less than the four-to-seven-point gain that motivated the project, and we report the measured value rather than a favourable subset. What the work does establish is a measurement (the noise is real, it is quantified per species and per pool, and it is reproducible from committed artefacts), a method (a frozen, unsupervised, auditable cleaning rule that needs no ornithologist), and a decision procedure for the remaining question (the three-arm control and the noise-scaling curve, both specified and implemented). The cleaned metadata and the reproducible loaders are released with the report so that the claim can be re-tested rather than taken on trust.

For future work we would extend the audit with a second, independent judge (for instance a second bioacoustic model or a small human-verified sample) to turn the binary triage into a graded confidence, extend the evaluation to five-fold cross-validation over recordings so that the test set is five times larger, and only then re-measure the accuracy difference with the confidence interval that the design deserves.

7. References

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8. Research Paper

A condensed IEEE-format paper based on this work is in preparation. Its core is the ablation of raw against audited-cleaned labels with the three-arm control and the noise-scaling curve; the artefacts needed to reproduce it (the audit, the cleaned metadata, the split and the scripts) are released with this report.

9. Curriculum Vitae

9.1 Madhur S. Mundra

B.Tech. in Information Technology, Dharmsinh Desai University, Nadiad (2023-2027). Enrolment no. 23ITUOZ066. Contact: madhur.mu5@gmail.com.

Project work: design and implementation of the unsupervised BirdNET audio audit, the two-stream 5.0 s chunking pipeline, the cleaned-metadata release and the reproducible MFCC+LSTM ablation described in this report.

Areas of interest: audio and speech processing, deep learning, data-centric machine learning, reproducible research.

9.2 Taksh N. Lade

B.Tech. in Information Technology, Dharmsinh Desai University, Nadiad (2023-2027). Enrolment no. 23ITUOS063. Contact: Takshlade982@gmail.com.

Project work: design and implementation of the unsupervised BirdNET audio audit, the two-stream 5.0 s chunking pipeline, the cleaned-metadata release and the reproducible MFCC+LSTM ablation described in this report.

Areas of interest: audio and speech processing, deep learning, data-centric machine learning, reproducible research.